Paleomagnetism of Archean rocks from northwestern Ontario: V. Poohbah Lake alkaline complex, Quetico Subprovince

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The Poohbah Lake alkaline complex is a late synkinematic Kenoran pluton in the Quetico gneiss belt of the western Superior Province. Three units of the complex, porphyritic syenite (PS), malignite (M: a nepheline–clinopyroxene–K-feldspar rock), and hornblende syenite (HS), as well as baked Archean schists near the intrusive contact, have a predominantly reverse-polarity R magnetization with mean direction $D = 198°$ $I = -22.5°$ ($k = 62$, $\alpha_{95} = 5°$, $N = 13$ sites) and a paleopole at 60°E, 50.5°N. Pyrrhotite and coarse primary magnetite are carriers of the R remanence. PS, M, and biotite pyroxenite (BP) exhibit also a predominantly normal N magnetization carried by fine-grained, probably secondary magnetite. N is systematically steeper than R: its mean direction is $D = 359.5°$ $I = +55.5°$ ($k = 28$, $\alpha_{95} = 8°$, $N = 13$ sites) with a paleopole at 90°E, 77.5°N. R and N do not record an asymmetric field reversal, since reverse-polarity N vectors and normal-polarity R vectors are occasionally found. R resembles in polarity and direction the natural remanent magnetization (NRM) of the 2630 Ma Matachewan diabase. It is probably the primary NRM of the Poohbah Lake pluton, with an age of about 2650 Ma in approximate agreement with the K/Ar isochron age of 2700 ± 25 Ma. N resembles in polarity and direction NRM's from the 2580 Ma Shelley Lake granite and the late Archean Burchell Lake granite. It is probably a secondary NRM about 2550 Ma in age, as suggested by updated K/Ar mica ages. The characteristic NRM of HS samples and secondary magnetizations in other rock types have poles on the Grenville Track of the polar wander path but there is no evidence for Grenvillian-age events in the area.

Le complexe alcalin du Poohbah Lake est un pluton syntectonique tardif kénorien de la ceinture gneissique Quetico dans le secteur ouest de la province du Supérieur. Trois unités du complexe, une syénite porphyritique (SP), une malignite (M: une roche à néphéline–clinopyroxène – feldspath K) et une syénite à hornblende (SH), et également des schistes archéens cuits au contact de l'intrusif, présentent une aimantation prédominante de polarité inverse (R) de direction moyenne $D = 198°$ $I = -22{,}5°$ ($k = 62$, $\alpha_{95} = 5°$, $N = 13$ sites) avec un paléopole à 60°E, 50,5°N. Les porteurs de la rémanence R sont la pyrrhotite et la magnétite primaire à gros grain. Les SP, M et pyroxénite à biotite (PB) exhibent également une aimantation normale N prédominante portée par de la magnétite à grain fin, probablement secondaire. L'aimantation normale N est systématiquement plus abrupte que R: sa direction moyenne est $D = 359{,}5°$ $I = 55{,}5°$ ($k = 28°$, $\alpha_{95} = 8°$, $N = 13$ sites) avec un paléopole à 90°E, 77,5°N. Les aimantations R et N n'indiquent pas de champ inverse asymétrique, vu que les vecteurs N de polarité inverse et les vecteurs sont décelés occasionnellement. La polarité et la direction de R ressemblent à l'aimantation rémanente naturelle (ARN) de la diabase de Matachewan âgée de 2630 Ma. C'est probablement l'ARN du pluton du Poohbah Lake, avec un âge d'environ 2650 Ma, correspondant approximativement à l'isochrone K/Ar de 2700 ± 25 Ma. La polarité et la direction de N ressemblent à l'ARN du granite du Shelley Lake âgé de 2580 Ma et du granite du Burchell Lake de l'Archéen supérieur. C'est probablement une ARN secondaire correspondant à l'âge de 2550 Ma, comme le suggèrent les âges K/Ar des micas. La particularité de l'ARN des échantillons de SH et des aimantations secondaires de d'autres types de roches est de présenter des pôles sur la courbe de la dérive apparente des pôles du Grenville, cependant il n'y a aucune indication d'événements d'âge grenvillien dans cette région.

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Introduction

Previous papers in this series have described the paleomagnetism and rock magnetism of late Archean mafic and granitic plutons of the Wabigoon, Quetico, and Shebandowan subprovinces in northwestern Ontario (Dunlop 1983*a*, 1984*a*,*b*; Dunlop *et al.* 1984). The present paper deals with similar studies carried out on the Poohbah Lake alkaline complex of the Quetico gneiss belt. A location map is given in Dunlop (1984*a*).

The Poohbah Lake Complex is an elliptical pluton that intrudes metasedimentary schists of the Couchiching Series (Fig. 1). Four lithologic units are studied in this paper: (1) porphyritic syenite, a nepheline-poor rock with phenocrysts of potassium feldspar in a groundmass of biotite, clinopyroxene, and blue–green alkali amphibole; (2) malignite, an equigranular nepheline–clinopyroxene – potassium feldspar rock with abundant apatite; (3) hornblende syenite; and (4) biotite pyroxenite. The individual units, including baked-contact rocks, have distinctive natural remanent magnetization (NRM) properties as a result of their rather different proportions of coarse- and fine-grained pyrrhotite, magnetite, and hematite.

There are two characteristic components of NRM. The reverse or R component is a south-southwest, shallow, upward magnetization carried mainly by pyrrhotite and coarse-grained magnetite. The normal or N component is a northward, intermediate, downward magnetization carried by fine-grained magnetite. The R component has the same direction as the NRM of the 2630 Ma Matachewan diabase (Irving and Naldrett 1977), whereas the N component resembles the primary NRM of the 2580 Ma Shelley Lake granite (Dunlop 1984*a*).

Geology and geochronology

The geology and K/Ar geochronology of the Poohbah Lake complex have been outlined by Mitchell (1976). These rocks contain primary biotites and amphiboles, and apart from some chloritization of biotites show little evidence of regional metamorphism. By contrast, the surrounding Couchiching schists have experienced amphibolite-grade metamorphism and considerable deformation; the final phase is due to intrusion of the alkaline complex. The Poohbah Lake pluton is therefore considered to be very late synkinematic, having been emplaced

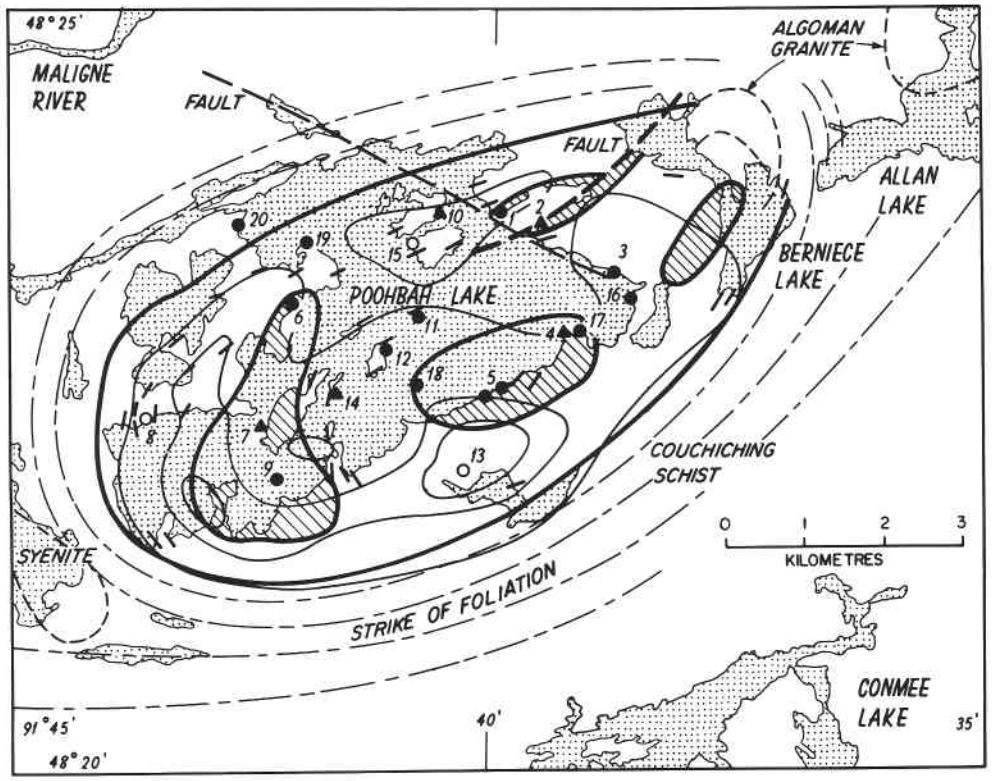


FIG. 1. Geological location map of the Poohbah Lake Complex and nearby syenites and Algoman granites, after Mitchell (1976). The heavy line marks the boundary of the pluton. Areas of predominantly malignite (M) outcrop are hatched; other areas are predominantly porphyritic syenite (PS). Short bars indicate feldspar phenocryst orientations in PS samples; these, like foliation trends in the Couchiching schists, follow the contours of the intrusion. Aeromagnetic anomaly contours (contour interval 100 nT) are shown as lighter lines. The anomaly peaks lie south and slightly west of the main M outcrops. Sampling sites are numbered. Solid circles are sites that yielded stable R and (or) N vectors; open circles are sites that yielded stable but unusual vectors; triangles are sites that yielded unstable results or only a few R or N vectors. The west-northwest-trending Hoffman Lake Fault passes near sites 1, 2, and 3.

during the waning stages of the Kenoran Orogeny.

Biotites from eight porphyritic syenite and malignite samples, as well as muscovites and biotites from three nearby granite and granite pegmatite bodies, gave apparent K/Ar ages between 2510 and 2630 Ma but an isochron age of 2700 ± 25 Ma (Mitchell 1976). Two other Poohbah Lake samples gave anomalously high apparent ages (2705 and 2760 Ma) and did not fit the isochron. Mitchell's interpretation is that the Poohbah Lake pluton and surrounding Algoman granites originally cooled around 2700 Ma but were disturbed by a later thermal event around 2550 Ma. Cross-cutting relationships show that the granites are younger than the alkaline complex, but the age difference cannot be resolved isotopically.

Mitchell (1976) explained the negative ^{40}Ar intercept of the K/Ar isochron by Ar loss during the 2550 Ma event, which he associated with minor strike-slip faulting. Sites 1 and 3 of the present study were on the northeast shore of Poohbah Lake close to the Hoffman Lake Fault, and site 2 was near a conjugate fault (Fig. 1). Paleomagnetic results from these sites show no noticeable enhancement of the N component, whose direction is consistent with magnetization in a 2550 Ma event. Faulting is therefore probably only one manifestation of a low-grade regional event of this age.

Sampling and measurement procedures

Seventy-five oriented block samples were collected during two field seasons from 20 sites in the Poohbah Lake Complex (Fig. 1). Sites 1, 3, 8, 10–13, 16, 17, and 19 were in porphyritic syenite (PS). (The site 1 rocks were equigranular but otherwise similar to porphyritic samples.) Sites 2, 5–7, 9, 14, and 18 were in malignite (M). Site 15 was in hornblende syenite (HS), site 4 was in biotite pyroxenite (BP), and site 20 was in baked Couchiching schist (CS) near the margin of the intrusion. Samples were oriented by magnetic compass, and strike readings were checked 1–2 m above the outcrop.

Three 2.5 cm cylindrical specimens of each sample were alternating field (AF) demagnetized in 12 steps to 1000 Oe (100 mT) using a Schonstedt single-axis demagnetizer. One specimen of each sample was thermally demagnetized in 50°C steps to 500°C and then in smaller steps to 675°C. The first season's collection was heated in a homebuilt noninductive furnace with Helmholtz coil field cancellation. Later samples were heated in a Schonstedt shielded furnace. Magnetizations were measured with Digico and Schonstedt spinner magnetometers.

Results were accepted if (1) specimen magnetization vectors reached a stable grouping during cleaning with <20° between-specimen divergence, or (2) if the magnetization vector swung along a great-circle arc during cleaning, even if a stable end point was not reached.

Thermomagnetic curves were measured in air for 23 representative samples using a recording Curie balance. The field applied was 3000–5000 Oe (300–500 mT). Initial susceptibility and saturation remanence were measured for 19 samples.

Thin-section studies

Polished thin sections of 11 samples were examined by L. D. Schutts. In transmitted light, PS samples (nine sections) were found to be equigranular to porphyritic amphibole syenites, consisting of perthitic orthoclase and microcline phenocrysts in a matrix of blue–green alkali amphibole, brown and green (partially chloritized) biotite, and some clinopyroxene and albite. Little nepheline was seen but sphene was abundant. Allanite and carbonates such as siderite were occasionally seen. M samples (two sections) were described as equigranular nepheline syenites with brown–green biotite, uralitized clinopyroxene, green amphibole, orthoclase, and abundant apatite and sphene.

In reflected light, both PS and M samples had as their most prominent opaques coarse (20–200 μm) to fine (<10 μm) primary magnetite or titanomagnetite grains only slightly subdivided by narrow lamellae in (111) planes. Grain boundaries were slightly to extensively hematized. Virtually all PS sections contained a few fine pyrrhotite grains, but pyrrhotite was not seen in M sections.

Thermomagnetic results

In thermomagnetic analysis, heating and cooling curves were practically reversible. Almost pure magnetite with a Curie temperature of 560–580°C was the main magnetic phase detected. There were minor inflections in the heating curves of some PS samples around 250°C but no other indication of pyrrhotite. Pyrrhotite is, however, clearly expressed in NRM blocking temperature spectra. BP samples, which contain much less magnetite than PS and M samples, have prominent contributions from paramagnetic silicates.

The probable reason for the subdued expression of pyrrhotite in strong-field magnetization is the following. Pyrrhotite has a critical single-domain (SD) size d_0 of 2–3 μm, whereas magnetite has $d_0 = 0.05$ μm (Dunlop 1981, Table 1). Thus micron-size and smaller pyrrhotite grains are SD, whereas the micron-size and larger magnetites seen in thin section are multidomain (MD) or pseudo-SD. MD grains contribute relatively much more to strong-field induced magnetization than to remanence, whereas the opposite is true of SD grains. The saturation magnetization of pyrrhotite is also only about one fifth that of magnetite.

NRM intensities and aeromagnetic interpretation

The Poohbah Lake body is marked by a positive aeromagnetic anomaly, whose peaks match areas of malignite outcrop but are displaced 0.5–1 km to the south (compare, for example, the M zone around sites 5 and 18 in Fig. 1 and the anomaly peak centred on site 13). In hand samples, remanent magnetization greatly outweighs induced magnetization (Koenigsberger ratio Q_n of 2–20), and the N component has an average intensity several times that of the R component.

The anomaly is therefore not due to magnetization induced by the steeply dipping (+72°) present Earth's field (PEF) but to the N remanence, which has an intermediate northward dip of about +55°. For this inclination, the anomalies detected by the airborne magnetometer should be offset to the south from surface sources by about 0.3 km, the aircraft terrain clearance. The greater offsets observed imply that the M zones extend to significant depths.

Surprisingly, however, NRM intensities have similar distributions and mean values in PS, M, and HS samples. The distributions are bimodal, with strong peaks between 2×10^{-5} and 10^{-4} emu/cm^3 (0.02–0.1 A/m) and between 2×10^{-4} and 10^{-3} emu/cm^3 (0.2–1 A/m). The main M anomaly source thus seems to be subsurface. BP and CS samples are rather weakly magnetized. Most have NRM's between 10^{-6} and 10^{-5} emu/cm^3 (10^{-3}–10^{-2} A/m).

Intensities of single-component NRM's tend to be 2–7% of saturation remanence. Laboratory thermoremanent magnetizations (TRM's) produced in a 0.5 Oe (0.05 mT) field also fall in this range. On this evidence, the NRM could be a primary TRM or a secondary thermal overprint, although a chemical remanent magnetization (CRM) is not ruled out.

N and R characteristic magnetizations

Porphyritic syenites

Figure 2 illustrates an interesting pattern common to most PS and a few M samples. On thermal demagnetization, the NRM vector swings towards a northerly, intermediate, downward direction. Frequently this N (normal-polarity) direction is reached as a stable end point between 350 and 550°C. The R or reverse-polarity vector removed below 350°C tends to fall in the southwest quadrant and to have a shallow, upward inclination.

On AF demagnetization, however, the resultant NRM vector swings in the opposite direction. The vector removed at low AF's is usually N, and an R end point is approached at high AF's. In a few cases (e.g., PO16, 27), true intermediate-direction vectors seem to exist, but the typical pattern is a simple N + R superposition.

Sample PO1 (Fig. 3) has an NRM that is almost purely R. Specimen 1/1 remains at the R direction from 0 to 1000 Oe (0–100 mT), 1/2 exhibits a short swing from the N direction from 0 to 250 Oe (0–25 mT), and 1/4 swings to an N direction above 300°C but the N vector is only a few percent of the original NRM. It is obvious from both total intensity and orthogonal vector plots that >90% of the R remanence has blocking temperatures between 300 and 350°C. The R cannot be a viscous remanence residing in magnetite because it has no blocking temperatures below 250°C and no coercivities below 1000 Oe (100 mT). Titanomagnetite with $x = 0.4$ would have an appropriate Curie point, but SD pyrrhotite is the likeliest carrier since pyrrhotite of this size was observed in most PS thin sections.

The predominant NRM component of PO2 (Fig. 4) is N. There is a minor viscous remanence that is eliminated by 300 Oe (30 mT) in AF cleaning or by heating to 304°C (see orthogonal vector plot). About 15% of the NRM is an R vector, with blocking temperatures mostly between 304 and 353°C but coercivities of >1000 Oe (100 mT). The remaining N remanence is carried by magnetite with blocking temperatures mainly between 550 and 575°C and coercivities distributed from 50 to at least 1000 Oe (5–100 mT). The AF intensity decay curve has SD form (Bailey and Dunlop 1983) and a median destructive field (MDF) of 550 Oe (55 mT). Such a high MDF in magnetite is impossible to explain by any pseudo-SD mechanism. Truly SD magnetite must be the source.

PO11 (Figs. 2, 5) has a more typical mix of R and N. The blocking temperature spectrum is partitioned between a probable viscous remanence (100–304°C), a pyrrhotite R vector (304–353°C) and a magnetite N vector (353–575°C). On AF demagnetization, only the N vector is clearly revealed. Judging by the AF intensity decay curve, the N is carried in this sample

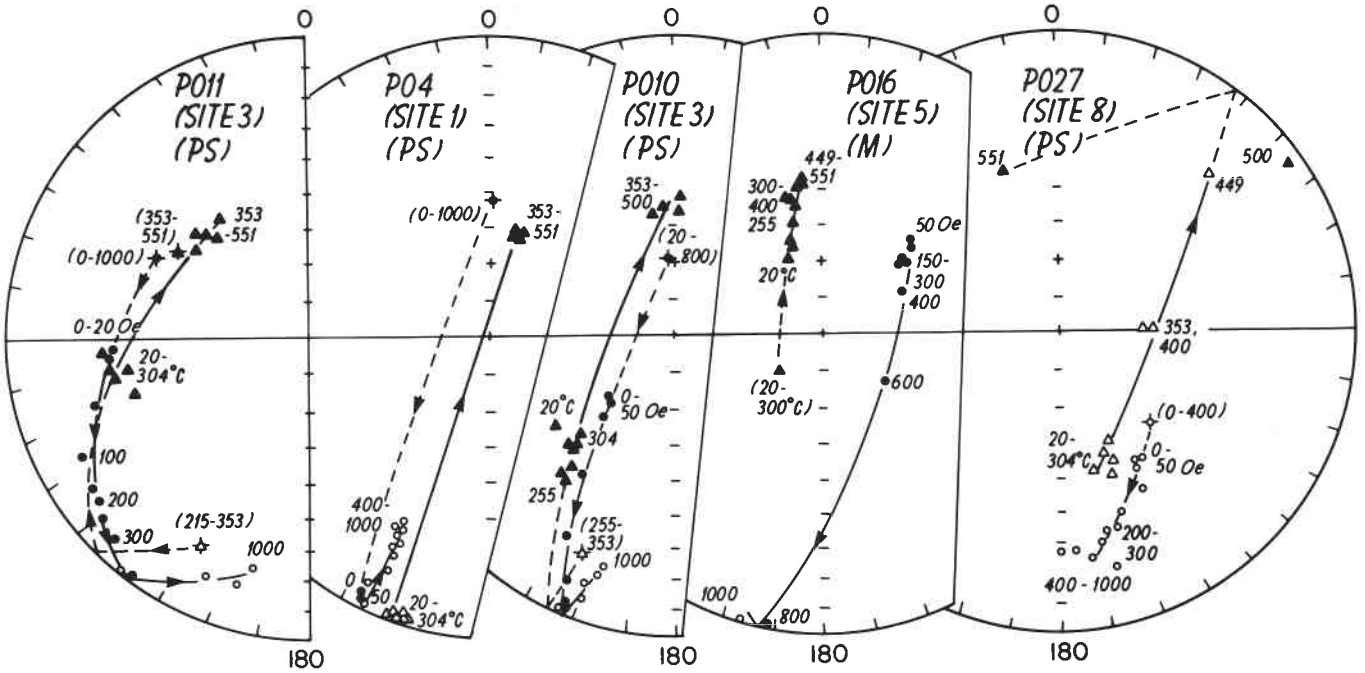


FIG. 2. Equal-area plots showing typical movements of NRM vectors towards the R direction on AF demagnetization (circles) and away from R on thermal demagnetization (triangles). Open and solid symbols denote upward and downward vectors, respectively. Complete R → N and N → R swings are seen in PO4, 10, and 11 (and in many samples not illustrated) when initial subtracted vectors (symbols marked with crosses) are taken into account. The N vector is usually 10−20° shallower than the PEF direction, whose approximate location is given by a cross.

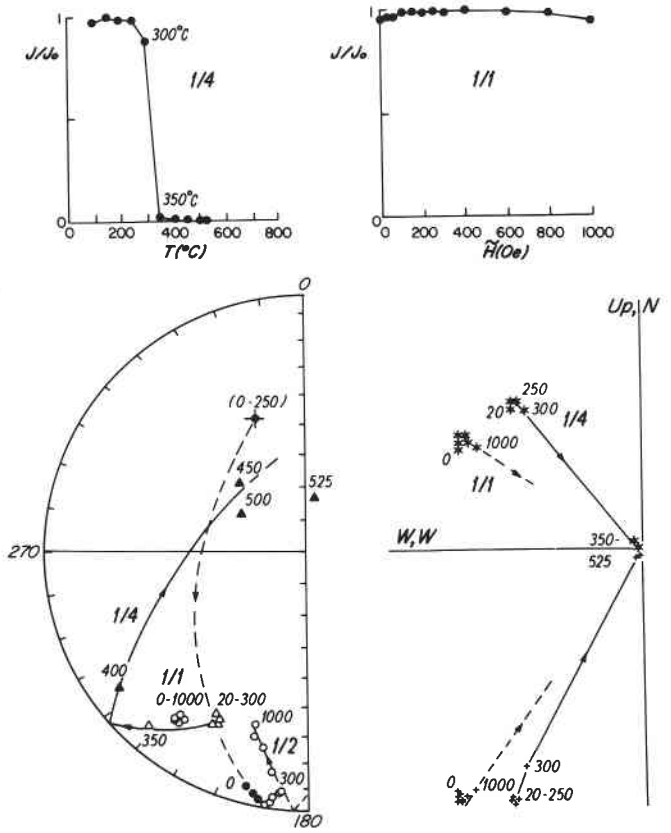


FIG. 3. Total intensity decay curves, equal-area plot, and orthogonal vector plots of AF and thermal demagnetization behaviour of porphyritic syenite sample PO1 (site 1), whose predominant magnetic phase is pyrrhotite. Crosses and stars on the vector plots denote horizontal-plane and vertical-plane projections, respectively. Other symbols as in Fig. 2.

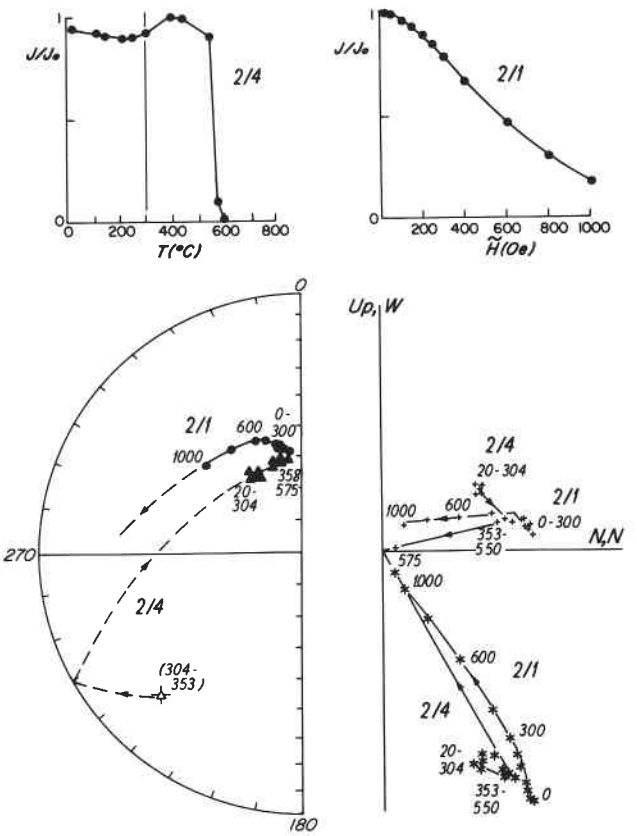


FIG. 4. AF and thermal demagnetization behaviour of porphyritic syenite sample PO2 (site 1), whose predominant magnetic phase is fine-grained magnetite with discrete blocking temperatures between 550 and 575°C. A little pyrrhotite with blocking temperatures of 300−350°C is present and carries a small R component.

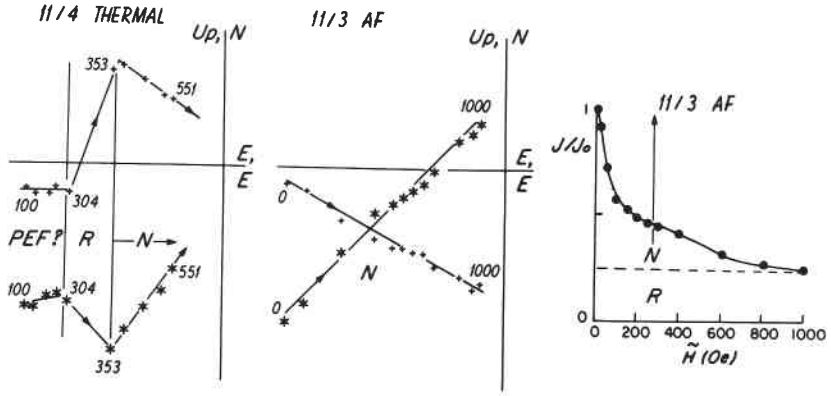


FIG. 5. Intensity decay and orthogonal vector plots of demagnetization behaviour in porphyritic syenite sample PO11 (site 3), whose NRM is bivectorial (N + R). Directional results were given in Fig. 2. The magnetite carrying N has both soft and hard fractions.

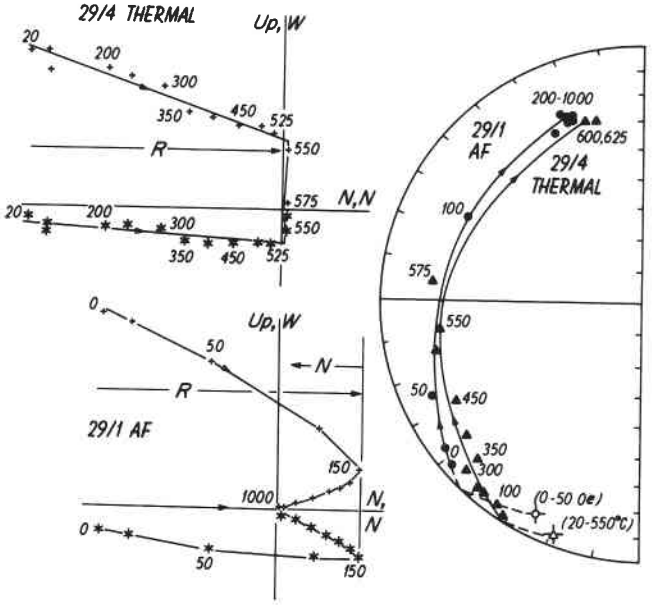


FIG. 6. Demagnetization behaviour of resultant magnetization vectors and orthogonal vector components in malignite sample PO29 (site 9). R is carried by soft, MD magnetite with T_B's up to 540°C and N is carried by hard, fine-grained magnetite with discrete, high T_B's.

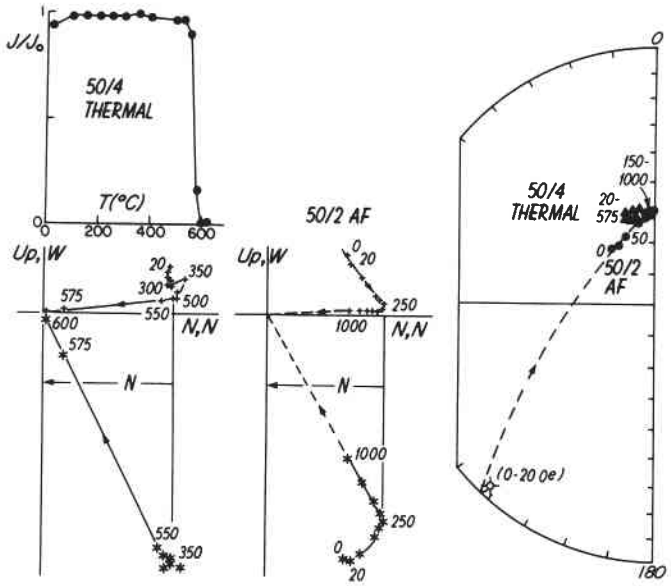


FIG. 7. Dominant N magnetization in malignite sample PO50 (site 14) carried by very hard magnetite with T_B's almost exclusively between 550 and 575°C. There is also a very small, relatively soft R component, whose H_C spectrum overlaps that of N.

by a bimodal mixture of fine SD magnetite like that in PO2, whose coercivity spectrum is seen above 250 Oe (25 mT), and coarse MD magnetite with an exponentially decaying AF spectrum seen below 250 Oe (25 mT). Another possibility is that the soft coercivities are associated with a viscous component like that seen in specimen 11/4, whose direction happens to be very close to that of N.

To summarize, porphyritic syenites have two characteristic remanences, separated in direction by about 150°. The R vector is carried by SD pyrrhotite with discrete T_B's, mainly between 300 and 350°C, and coercivities usually >1000 Oe (100 mT). It is isolated as a stable end point by AF cleaning. The N vector is carried by SD and perhaps also by MD magnetite. Its T_B spectrum is usually discrete and concentrated between 525 and 575°C but is sometimes distributed to lower temperatures. The coercivity (H_C) spectrum is typically SD in form but low H_C's (<250 Oe (25 mT)) are sometimes also found. The low H_C's and T_B's may, however, be associated with viscous remanences whose directions are close to N. The N vector is isolated as a stable thermal end point.

Malignites

Malignites have the same R and N components as porphyritic syenites, but the R vector is carried mainly by magnetite rather than pyrrhotite. PO29 (Fig. 6) is a typical example. The R component is revealed by subtracted vectors in both AF and thermal cleaning. Its coercivity spectrum is very soft (AF vector plot): most H_C's fall between 20 and 150 Oe (2 and 15 mT), but T_B's extend from 150 to ~540°C. These H_C and T_B ranges are characteristic of coarse MD magnetite (Dunlop 1983*b*, 1984*a*; Dunlop *et al.* 1984). The smaller N vector has T_B's concentrated between 540 and 575°C and H_C's from 150 to 1000 Oe (15−100 mT). It is probably carried by SD magnetite.

In sample PO50 (Fig. 7), there is no sign of an R component in the directional data during thermal or AF cleaning. However, a small, soft R vector probably carried by MD magnetite is revealed as a subtracted vector at low AF's in specimen 50/2 (AF vector plot), and a tiny pyrrhotite component seen in the vector plot of 50/4 between 300 and 350°C is probably also R. The predominant N component is carried by very hard, presumably SD magnetite with H_C's of 250 to >1000 Oe (25−>100 mT) and discrete T_B's about 80% of which are

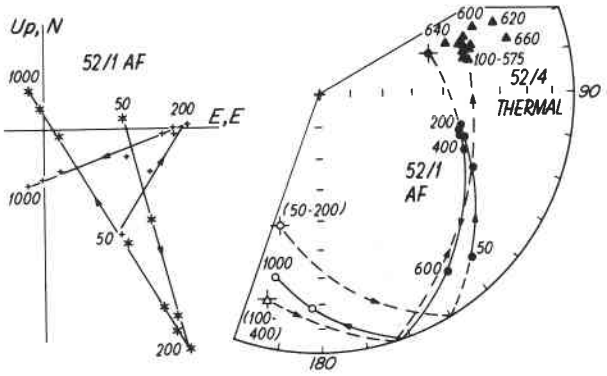


FIG. 8. Superimposed low- and high-coercivity R remanence and an intermediate-coercivity HS characteristic remanence in hornblende syenite sample PO52 (site 15).

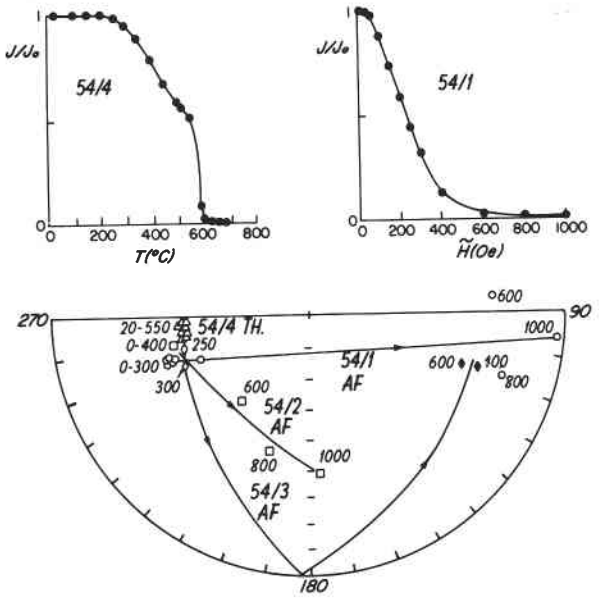


FIG. 9. Superimposed normal and reverse-polarity HS characteristic remanences in hornblende syenite sample PO54 (site 15).

between 550 and 575°C. The unusually high coercivities cannot be attributed to hematite, since there are virtually no T_B's > 600°C.

In summary, malignites exhibit the same R and N remanences as PS samples, but N is usually the only characteristic remanence isolated at high AF's or temperatures. N is carried by very hard SD magnetite and R by soft MD magnetite and to a minor extent probably also by pyrrhotite.

Biotite pyroxenites

Biotite pyroxenites from site 14 are weakly magnetized but have simple behaviour. On thermal and AF demagnetization, they swing from a steep-inclination direction near the PEF to an intermediate-inclination N end point. The PEF viscous remanence has T_B's mainly between 100 and 250°C and coercivities below about 200 Oe (20 mT). The N vector has discrete T_B's between 550 and 575°C and hard H_C's distributed from about 100 to >1000 Oe (10−>100 mT). It is carried by SD magnetite with properties similar to the N carrier in M samples.

Hornblende syenites

Hornblende syenites from site 15 have unique properties. Some have R vectors but none exhibits the N component.

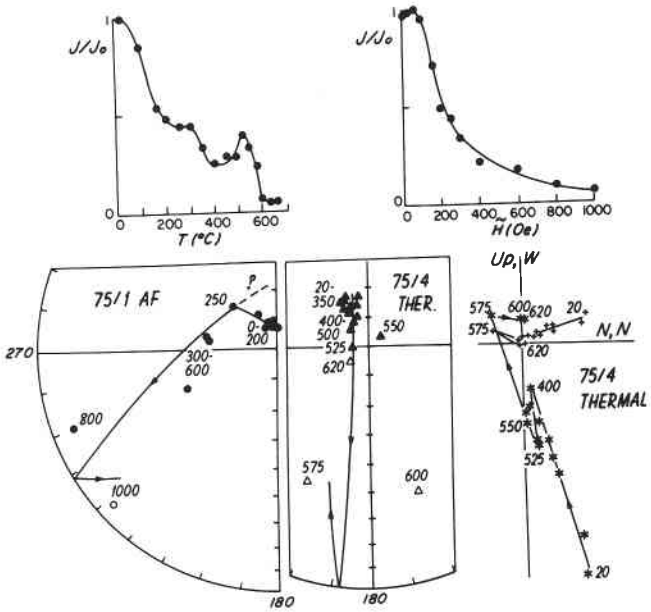


FIG. 10. PEF (+N?) + R magnetizations in baked Couchiching schist sample PO75 (site 20, contact test). The R vector is carried by hematite. Unbaked CS samples carry only unstable remanence.

Instead they display a dual-polarity characteristic remanence with easterly or westerly declination and intermediate inclination. PO52 (Fig. 8), for example, contains in its NRM an R vector with low T_B's (100−400°C) but both low and high H_C's (50−200 and >1000 Oe (5−20 and >100 mT)). The R component is probably carried by both MD magnetite and SD pyrrhotite, although the low T_B's are hard to account for in the former case.

The HS characteristic remanence has coercivities from 200 to 1000 Oe (20−100 mT) (since the resultant vector has reached the R direction at 1000 Oe (100 mT), the HS characteristic must be essentially destroyed) and blocking temperatures from 400 to at least 660°C. The carriers are SD magnetite and minor hematite.

Figure 9 illustrates an almost complete 180° reversal of the HS characteristic remanence in sample PO54. The west-declination upward vector is predominant. It has both distributed T_B's (300−500°C) and discrete T_B's (550−575°C), likely caused by magnetite in two different size fractions. The coercivity spectrum is softer than in PS, M, and BP samples: the magnetite is probably pseudo-SD rather than truly SD (<1 μm but >0.05 μm in size). The reversals occur at high AF's when only a few percent of the NRM remains. The vector swings are widely spread in azimuth, a hallmark of 180° reversals.

Couchiching schists (contact test)

CS samples at site 20 are considerably less than one intrusion width from the contact (Fig. 1) and must have been baked by the Poohbah Lake intrusion. They yield an indeterminate contact test: they exhibit the same R remanence found in the intrusion and probably also the N remanence, but they do not appear to retain any distinctive pre-intrusion remanence of their own. CS samples taken along the Maligne River far from the intrusion are weakly and unstably magnetized and give no hint of what the CS characteristic direction might be.

Figure 10 shows some representative results. From 0 to 200 Oe (0−20 mT), a steep vector representing more than one half the NRM is erased. This remanence was probably acquired in the PEF. From 250 to 1000 Oe (25−100 mT), the resultant

swings from what is likely the N vector towards R. The thermal behaviour is less easily interpreted, since there is only one obvious great-circle swing. Two components with similar directions are erased between 20 and 400°C and between 525 and 575°C, respectively, with a curious antiparallel vector perhaps reflecting laboratory chemical change being uncovered between 400 and 525°C. The first component is probably viscous and the second is probably the N vector, with the usual discrete magnetite blocking temperatures. A significant R component carried by hematite is revealed between 575 and 620°C, explaining the high H_C's of R in AF cleaning.

There are several possible scenarios that would explain the CS results.

(1) The R vector in baked CS rocks is carried by primary hematite and is a thermal overprint caused by reheating by the Poohbah Lake intrusion. The N is a CRM carried by secondary magnetite generated in a later (2550 Ma?) mild thermal event. The PEF has overprinted the finer magnetite grains. In this model, R is older than N, and the R component in Poohbah Lake rocks is a primary TRM.

(2) The N vector in baked CS rocks is the intrusion thermal overprint and R is a CRM carried by secondary hematite. In this model, N predates R, and in the Poohbah Lake body N is the primary TRM.

(3) There was a single regional remagnetizing event subsequent to intrusion of the Poohbah Lake Complex. Slow cooling of both CS and the Poohbah Lake rocks after this event would account for the rather different R and N directions. Note that R and N cannot both be thermal remanences acquired in a single cooling event, since they have adjacent or overlapping T_B's but rather different ages (they are noncoaxial by >30°). One or the other must be a CRM. On this model, the relative ages of R and N are ambiguous and neither component is the primary remanence of the Poohbah Lake pluton.

Ages of the R and N remanences

The CS contact test fails to prove that either N or R is the Poohbah Lake primary remanence. Nor is it obvious which of the three suggested scenarios is the true one. However, the *directions* of the R and N vectors place constraints on their ages. As we shall see later, the paleomagnetic pole corresponding to R is consistent with a magnetization age of about 2630 Ma, whereas the N paleopole suggests a younger age of 2550−2600 Ma. We shall now review the PS, M, BP, and HS data to see if they are consistent with the age sequence suggested by the paleopoles.

The R vector is found in most PS and M and some HS samples. In PS rocks, R is carried exclusively by pyrrhotite and disappears above the Curie temperature of about 320°C. In M samples, R is carried principally by soft, primary MD magnetite with T_B's as high as 540°C. In HS samples, both pyrrhotite and MD magnetite likely carry the R component.

The N vector is found in most PS, M, and BP samples. It is carried by fine SD and (or) PSD magnetite, of a grain size too fine to be seen optically. In addition, MD magnetite *may* in a few cases carry a low-H_C, low-T_B N remanence (e.g., Fig. 5) but it is equally plausible that this magnetization is a PEF overprint. Since N characteristically has T_B's of 500−575°C that overlap those of the R remanence carried by MD magnetite, N and R cannot both be totally thermal in origin.

The only model that explains both blocking temperatures and paleopoles is scenario 1. The pyrrhotite (both visible and submicroscopic) and coarse-grained MD magnetite are primary. They acquired the R remanence as a primary TRM in initial cooling of the Poohbah Lake pluton 2600−2700 Ma ago, the event recorded by the K/Ar isochron. The contact rocks contained primary hematite, which upon heating had its NRM reset to the R direction. After the geomagnetic field had reversed polarity and appreciable APW (apparent polar wander) had occurred 2550−2600 Ma ago (updated K/Ar biotite ages) there was a mild event of regional extent that imprinted the N remanence on all rock types, including CS. The N magnetization is a CRM carried by fine-grained secondary magnetite produced during this event. Since R blocking temperatures are sometimes as low as 100−150°C (e.g., Fig. 8), the secondary magnetite was probably produced at quite low temperatures.

Scenario 2 is unlikely. It not only violates the age sequence suggested by the paleopoles but also implies that the coarse MD magnetite carrying the R vector in malignites is secondary. Viewed in thin section, these oxides are definitely primary. They could have been overprinted thermally but not chemically. Scenario 3 is possible but less likely than 1, since the very different T_B's of R in PS (300−350°C), M (150−540°C), and CS (575−620°C) seem to require rapid cooling.

Scenario 1 should be regarded as hypothetical until it has been tested by isolating the source of the submicroscopic magnetite that carries N, by magnetic microanalysis (Buchan 1979), for example, and by verifying that CS rocks far from the contact carry only N and baked CS samples carry both R and N. In view of the inherent instability of unbaked CS samples, the latter will be no easy task.

Average directions of R and N magnetizations

End points and subtracted vectors

Cleaned sample-mean and site-mean characteristic magnetization directions are plotted in Fig. 11. A statistical analysis of R and N data separately is provided in Table 1. Outlying data from sites 7 (samples 23, 24) and 17 (samples 60, 61) were excluded in calculating the grand means.

Vectors from site 8 (samples 26−28) are of reverse polarity but have the N direction. In fact, during AF and thermal cleaning all four specimens of sample 26 exhibit complete 180° R → N reversals (Fig. 12) with a wide spread of great-circle azimuths. The site 8 results were not included in the R population for averaging.

HS characteristic directions of both polarities from site 15 (samples 52−55) represent a distinct population. They were averaged separately from the R and N populations. The corresponding HS paleopole will be discussed later.

The distribution of N vectors is not particularly Fisherian but the N grand mean, $D = 359.5°$ $I = +55.5°$ ($k = 28$, $\alpha_{95} = 8.3°$, $N = 13$ sites representing 35 samples), is very similar to the grand mean of type 1 NRM's in the nearby Shelley Lake granite (SL1, Dunlop 1984*a*; cf. Fig. 11), which did have a Fisherian distribution. SL1 was argued to be a primary TRM 2580 ± 20 Ma in age. The Poohbah Lake N magnetization (PL_N), although not primary according to scenario 1, is likely similar in age.

The R vectors have a Fisherian distribution with mean $D = 198°$ $I = -22.5°$ ($k = 62$, $\alpha_{95} = 5.3°$, $N = 13$ sites representing 39 samples). This mean is very similar to NRM's of certain Archean greenstones in the Wabigoon Subprovince (Dunlop 1983*a*) and to results from 2630 Ma Matachewan dikes in the Abitibi Subprovince (Irving and Naldrett 1977; Schutts and Dunlop 1981). The Poohbah Lake R NRM (PL_R) is probably of similar age.

TABLE 1. Poohbah Lake Complex, site-mean magnetization directions based on AF and thermally cleaned results (end points and subtracted vectors)

Site	Rock type	N	D (°)	I (°)	k	α_{95} (°)
1	PS	3	211	−27.5	62	16
2	M	1	210	−24		
3	PS	4	193	−24	1094	2.5
5	M	2	198	−14.5	113	
7*	M	2	155.5	−11	74	
8*	PS	3	176	−63.5	31	22.5
9	M	4	190	−14	38	15
11	PS	4	206	−27.5	118	8.5
12	PS	3	200	−12	158	10
13	PS	1	184.5	−11.5		
15	HS	1	196	−26		
16	PS	3	192.5	−23.5	990	4
17	PS	4	199	−35.5	26	18
18	M	1	203	−20		
19	PS	3	193	−28.5	71	15
Mean, unit weight to sites		13*	198	−22.5	62	5.3
Paleopole			60E	50.5N	($\delta p = 3°$, $\delta m = 5.5°$)	
1	PS	3	346	58.5	30	23
3	PS	3	320.5	59	18	30
4	BP	2	348	47.5	57	
5	M	2	12	52.5	334	
6	M	4	25.5	48	20	21
7	M	2	29	42	14	
8	PS	1	24	58		
10	PS	1	350	63		
12	PS	1	0	55.5		
14	M	2	359	59	260	
15*	HS	4	72	51.5	23	19
16	PS	2	321	49.5	23	
17*	PS	2	332	18.5	65	
18	M	4	345	53	97	9
19	PS	2	21	53.5	20	
Mean, unit weight to sites		13*	359.5	55.5	28	8.3
Paleopole			90E	77.5N	($\delta p = 8.5°$, $\delta m = 12°$)	

NOTES: N is the number of samples or sites averaged; D, I are declination and inclination of magnetization; k is the Fisher precision parameter; and α_{95} is the radius of the 95% confidence cone about the mean direction. Reverse (R) and normal (N) polarity results are listed and averaged separately. Key to rock types: PS = porphyritic syenite; M = malignite; HS = hornblende syenite; BP = biotite pyroxenite.

*Results for sites 7 and 8 not included in R mean; results for sites 15 and 17 not included in N mean.

Intersecting remagnetization circles

Average R and N directions were also estimated by the method of converging remagnetization circles (Halls 1976, 1978). Since each remagnetization circle incorporates both R and N data, convergence points tend to be biassed toward a direction intermediate between R and N (B. J. J. Embleton, personal communication, 1984). The approach taken here was to consider separately subpopulations in which R and N vectors, respectively, dominate.

For example, great circles intersecting near the R direction are illustrated in Fig. 13*a*. The best estimate of the R intersection point (Fig. 13*b*; Table 2) is slightly steeper than the grand mean of stable end-point directions, as a result of the bias mentioned above, but is not distinguishable from it at the 95% confidence level. The N intersection point (circles not illustrated) is similarly indistinguishable from the N grand mean.

The concentration of great-circle azimuths to the south-southwest is a result of the non-180° separation of R and N. Reversals of 180°, like those of Figs. 9 and 12, tend to have widely spread great-circle azimuths.

Secondary magnetizations

Low-T_B, low-H_C secondary magnetizations were separated from R and N characteristic remanences in 55 PS, M, and BP samples using orthogonal vector plots. About one half of the separated vectors are PEF overprints with northeast to northwest declinations and steep positive inclinations. Of the remainder, most are widely dispersed and extremely soft; they are probably viscous remanences acquired in the laboratory. There are, however, two groups of vectors for which there is some evidence of an ancient origin.

A possibly ancient secondary remanence in PS sample PO57 is illustrated in Fig. 14. The overprint vector, which has a northwest declination and intermediate negative inclination, is removed in part between 100 and 300°C. Between 300 and 350°C, a pyrrhotite R component is erased. Between 350 and

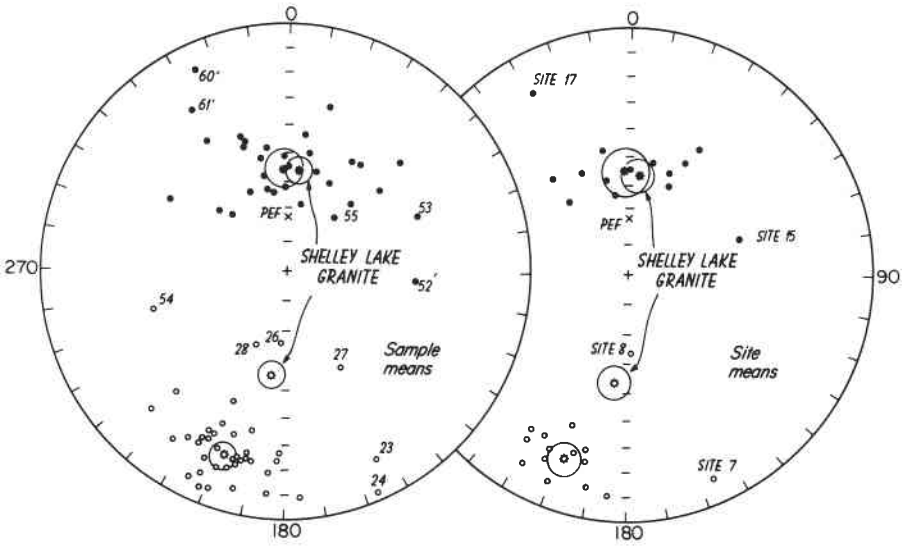


FIG. 11. Cleaned sample-mean and site-mean magnetization directions for samples from Poohbah Lake alkaline complex. Sample and site numbers are given for divergent stable results. Mean directions of R and N together with their circles of 95% confidence are compared with results from the Shelley Lake granite (Dunlop 1984*a*). The N means are distinct from the PEF direction at the 95% confidence level. HS characteristic magnetizations from site 15 (samples 52–55) form a distinct grouping similar to some groups of secondary magnetizations from other rock types.

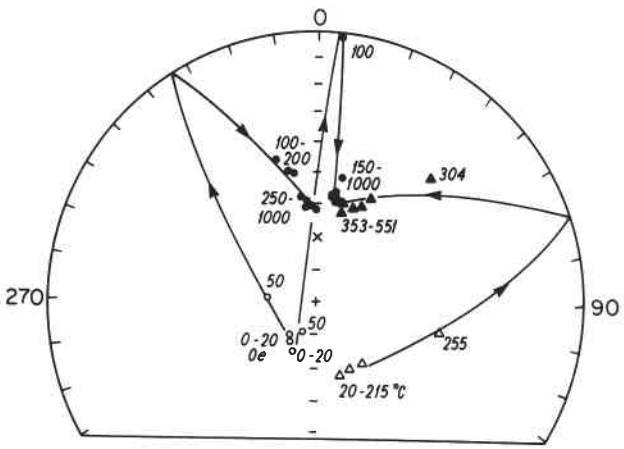


FIG. 12. The 180° reversals of magnetization in the course of AF and thermal cleaning of porphyritic syenite sample PO26 (site 8). The initial reverse magnetization is much steeper than the R vector seen at other sites.

TABLE 2. Poohbah Lake Complex, analysis of remagnetization circles intersecting near the R and N mean directions of Table 1

	N	D (°)	I (°)	k'	β_{95} (°)	γ_{95} (°)
R, unit weight to specimen normals	79	197	−33	32	3.5	5.5
Paleopole		58E	57N	($\delta p < 4°$, $\delta m < 6°$)		
N, unit weight to specimen normals	22	17	49	52	5	9.5
Paleopole		47E	67N	($\delta p < 9°$, $\delta m < 13°$)		

NOTES: The parameter k' corresponds to the Fisher k of Table 1 and the confidence oval semi-axes β_{95} and γ_{95} (Bailey and Halls 1984) are analogous to α_{95}.

525°C and between 50 and 200 Oe (5 and 20 mT), the T_B and H_C spectra of the overprint overlap those of the N vector. The overprint has T_B's well above 300°C, the maximum expected for viscous remanence acquired at ordinary temperatures by magnetite (Wilson and Smith 1968; Dunlop 1969; Pullaiah *et al.* 1975). It is probably a viscous partial TRM (Briden 1965) produced by a very mild thermal event.

Similar trivectorial remanences occur in other PS samples from sites 16 and 19 and in a single M sample, PO68 from site 18. The overprint directions resemble later Grenvillian (950–850 Ma) or Early Cambrian results for North America (cf. Irving 1979, Figs. 8, 9), although no event of either age is known to have occurred in this area. A roughly similar direction is recorded by the A remanence of rocks from the Kapuskasing Structural Zone about 650 km to the east, but Ernst and Halls (1984) interpreted the A to be structurally rotated.

The other secondary remanences that may have significance are exhibited mainly by M samples and group around $D = 90°$, $I = +45 - +75°$. This grouping and the HS characteristic remanence have directions consistent with an early Grenvillian (1050–950 Ma) age of magnetization.

Both groups of possibly ancient overprints are based on relatively few samples within the total population. The notion of Grenvillian overprinting is therefore highly speculative, particularly as there is no independent evidence of any "echo" of the Grenvillian Orogeny in the western Superior Province.

Paleopoles and paleolatitudes

Paleomagnetic poles PL_R and PL_N corresponding to the R and N characteristic remanences were calculated in Tables 1 and 2. In Fig. 15, they are compared with the North American APW path for the Archean. PL_R falls in a group formed by poles MD and MD2 of the 2630 Ma Matachewan dikes (Irving and Naldrett 1977; Schutts and Dunlop 1981) and KJ of the Kinojevis tholeiites (Geissman *et al.* 1983), all of the Abitibi Subprovince. PL_R also resembles closely the virtual geomagnetic pole (VGP) calculated for the Dyment greenstone (DY in Fig. 12 of Dunlop 1983*a*), which is 150 km north of Poohbah Lake in the Wabigoon Subprovince.

Pole PL_N resembles poles SL1 and BL1 from type 1 NRM's of the Shelley Lake granite, 25 km east of Poohbah Lake (Dunlop 1984*a*), and the Burchell Lake granite, 85 km to the

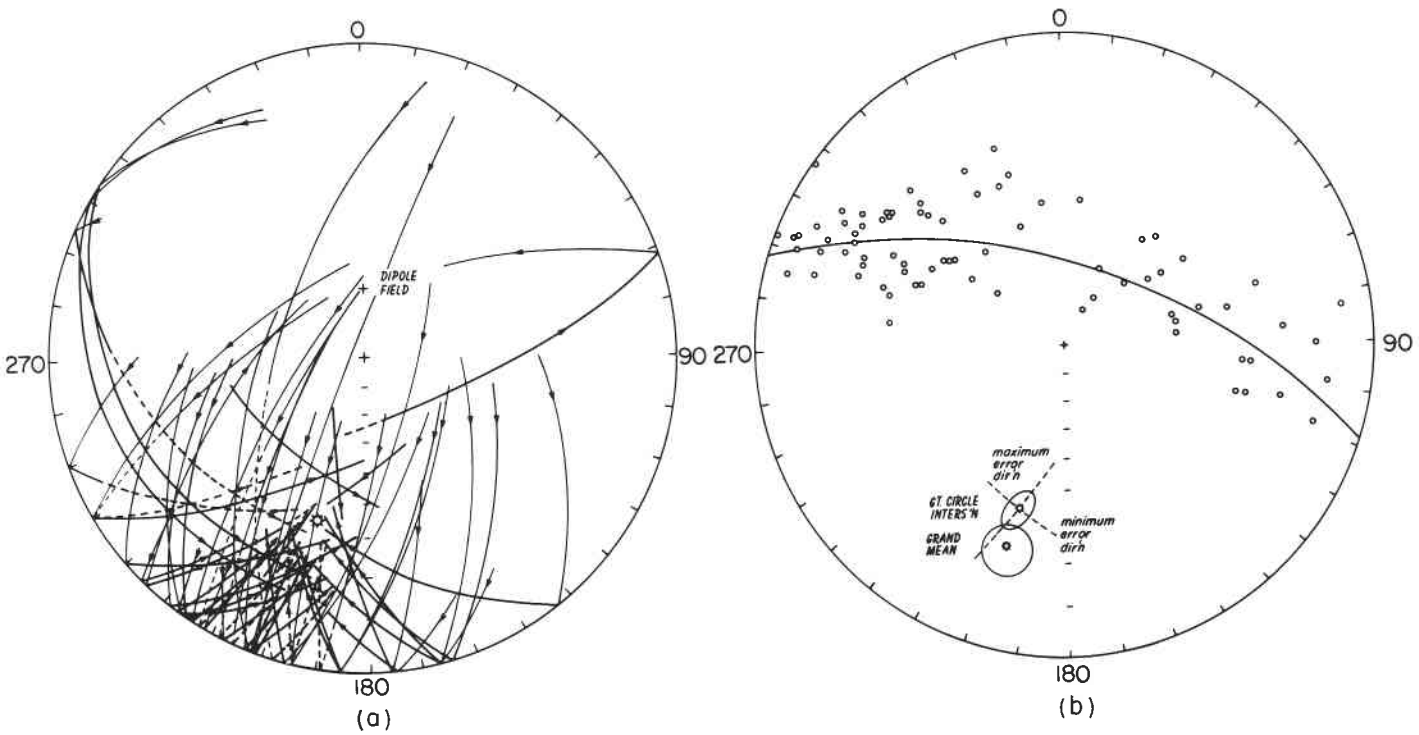


FIG. 13. (*a*) Stereographic projection of great-circle swings of resultant magnetization vectors (remagnetization circles) intersecting near the R direction. (*b*) Upward normals to the remagnetization planes defined by the circles in (*a*) and the best fitting great circle through the normals. The normal to the plane defined by this latter circle is the best estimate of the intersection point of the original remagnetization circles. It agrees reasonably well with the grand mean of Fig. 11.

east in the Shebandowan Subprovince (Dunlop 1984*b*). SL1 was argued to have a primary magnetization age of 2580 ± 20 Ma.

At the time PL_R was acquired, the part of ancestral North America containing the Superior craton was rotated about 20° counterclockwise compared with the present orientation and was located in equatorial latitudes (about 10°N). By the time of PL_N, the craton had drifted to the middle latitudes (about 35°N) and had rotated to its present azimuth.

The HS characteristic remanence yields a VGP (not shown) at 167°E, 35°S ($\delta p = 17.5°$, $\delta m = 26°$), on the pre-1000 Ma part of the Grenville Track of the APW path (Berger *et al.* 1979). The VGP's of the ancient secondary remanences discussed in the last section trend along the 1000–820 Ma part of the same track.

Discussion

Halls and Pesonen (1982) have suggested that during Keweenawan time (1150–1050 Ma) reversals of the geomagnetic field may have been asymmetric. Could the Poohbah Lake R and N remanences be the record of an asymmetric reversal in the late Archean? The answer is no. Although the N remanence is predominantly of normal polarity, samples at site 8 (Fig. 11) have antipodal NRM directions and specimens of PO26 (Fig. 12) exhibit 180° reversals between the two polarities. Similarly, there are a few examples of R vectors with normal polarity, including two complete 180° reversals of NRM in specimens of PO32 (M, site 9) during AF cleaning (not illustrated).

R and N are therefore spot readings of the field at two times sufficiently different for about 25° of APW to have occurred. If R was acquired around 2650 Ma, an estimate compatible with the age of the Matachewan diabase (about 2630 Ma) and the Poohbah Lake K/Ar isochron age of about 2700 Ma, and N was acquired around 2550 Ma, which agrees with the Shelley Lake granite magnetization age of 2580 ± 20 Ma and the updated Poohbah Lake K/Ar biotite ages, an APW rate of less than 30 mm/year is required. This rate is comparable to APW rates of 30–40 mm/year in the Late Proterozoic (e.g., 50° of APW between 980 and 820 Ma, Berger *et al.* 1979) and to present-day sea-floor spreading rates in the Pacific.

The pronounced polarity bias in both R and N remanences implies long periods of one polarity at this time in the Archean. It is notable that Matachewan diabase results are >90% reversed (Irving and Naldrett 1977; Ernst and Halls 1984) and 60–70% of Shelley Lake and Burchell Lake type 1 data are normal (Dunlop 1984*a*,*b*).

The 2500–2600 Ma event that produced the N remanence could have been mild regional heating associated with intrusion of Algoman granites throughout the Quetico Belt. If so, the Algoman and Kenoran "orogenies" must have been separate events rather than phases of the same event, since they have distinct paleomagnetic signatures. This hypothesis would explain why granitic plutons in the Quetico and Shebandowan belts carry only the N remanence, whereas basal volcanics and mafic dikes in the Wabigoon and Abitibi belts have only the R remanence. On this model, R would be a late synkinematic Kenoran remanence and N a postkinematic Algoman remanence. The fact that K/Ar data from Algoman granites and pegmatites near Poohbah Lake fit a 2700 Ma isochron speaks against this idea, but the only sure proof will be a paleomagnetic study demonstrating that these granites either possess or lack the R remanence.

Conclusions

(1) The Poohbah Lake Complex carries two characteristic

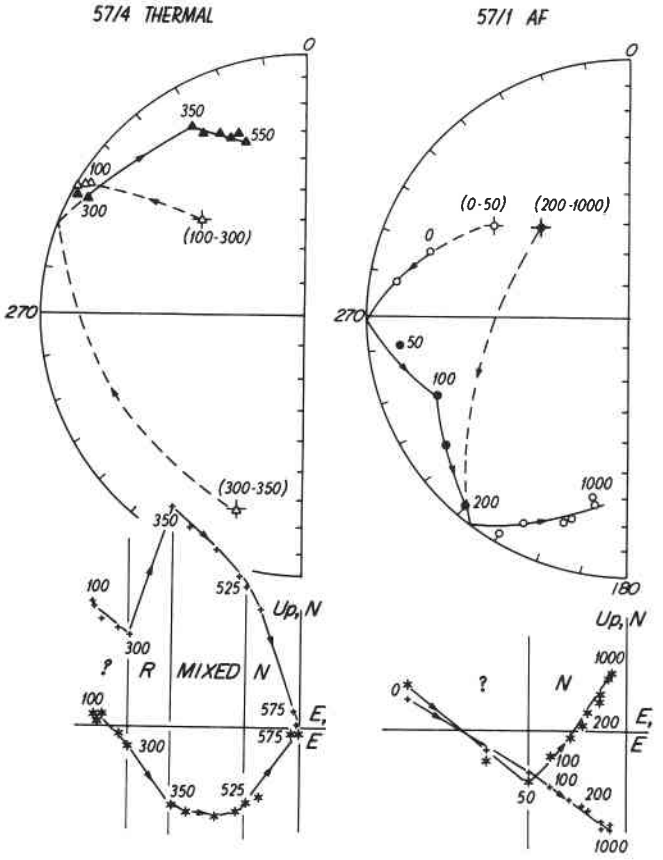


FIG. 14. Trivectorial magnetizations in porphyritic syenite sample PO57 (site 16). The initial low-T_B, low-H_C secondary magnetization is superimposed on R and N components. Its T_B spectrum overlaps that of N up to about 525°C.

remanences: R (predominantly reverse polarity) with a paleopole at 60°E, 50.5°N, and N (predominantly normal) with a paleopole at 90°E, 77.5°N.

(2) R and N do not record an asymmetric reversal of the geomagnetic field but spot readings at different times.

(3) The R remanence is carried by pyrrhotite in porphyritic syenites and by coarse primary magnetite in malignites. It is probably a primary TRM about 2650 Ma in age; its pole is close to that of the 2630 Ma Matachewan diabase.

(4) The N remanence is carried by fine, single-domain magnetite in porphyritic syenites, malignites, and biotite pyroxenites. It is probably a CRM about 2550 Ma in age; its pole is close to that of the 2580 Ma type 1 NRM of the Shelley Lake granite.

(5) Hornblende syenites in the complex have a characteristic remanence with a VGP at 167°E, 35°S on the early (>1000 Ma) part of the Grenville Track. Some secondary remanences in other rock types form groups with poles on younger parts of the Grenville Track. However, there is no independent evidence for an "echo" of the Grenvillian Orogeny in this part of the Superior Province.

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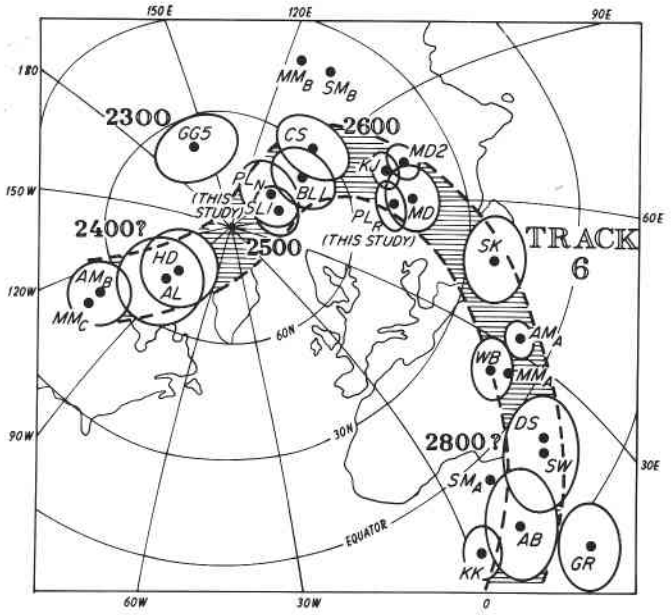


FIG. 15. Tentative late Archean APW path for Laurentia. The various primary and secondary paleopoles are identified in Irving (1979, Fig. 11) and Dunlop (1984*a*, Fig. 12). The positions of poles PL_R and PL_N of the present study are consistent with magnetization ages of ~2650 and ~2550 Ma, respectively. These ages in turn are supported by K/Ar isochron and updated mica ages (Mitchell 1976).

Larry Schutts performed the optical thin-section examinations, Nick Maffei determined Curie points, Christopher Hale measured susceptibilities, saturation remanences, and TRM, and Charlotte Jackson translated the intersecting remagnetization-circle program of Bailey and Halls (1984) into the microcomputer form used in this paper. To all of these people I am grateful. Useful reviews by K. L. Buchan, B. J. J. Embleton, R. J. Enkin, and Ö. Özdemir were much appreciated. This work was supported by Energy, Mines and Resources Canada (EMR) research agreements and by Natural Sciences and Engineering Research Council of Canada (NSERC) operating grant A7709.

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